

CURRICULUM VITAE – YUNFEI MA (U.S. PERMANENT RESIDENT)

SENIOR ENGINEERING MANAGER, MOBILE AND EDGE NETWORKING

PERSONAL INFORMATION Yunfei Ma
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CURRENT POSITION **Senior Engineering Manager**
Uber Technologies, Inc.
Leading Uber's Mobile and Edge Networking organization
Sunnyvale, CA

WORK EXPERIENCE **Uber Technologies** Sunnyvale, CA, Sept. 2023–present
Sr. Staff Engineer → Sr. Staff & TLM → Sr. Engineering Manager

Edge Networking

The Edge Networking team manages all Layer 7 traffic into Uber, supporting both mobile and web platforms. The team manages L7/L4 edge proxies across cloud and on-premises environments, as well as Uber's global DNS infrastructure, anti-DDoS infrastructure, and network security.

Mobile Networking

The Mobile Networking team owns the mobile networking stack, libraries, and observability infrastructure across all Uber apps on iOS and Android.

Autonomous Vehicle Networking

The Autonomous Vehicle Networking team enables reliable vehicle-to-cloud connectivity for autonomous vehicles and owns Uber's video streaming platform.

Alibaba Group (U.S.) Sunnyvale, CA, Sept. 2018–Aug. 2023
Senior Engineer → Staff Engineer → Sr. Engineering Manager

XLINK: Led the design and development of the industry's first widely deployed multi-path QUIC stack, using a cross-layer approach to overcome traditional multipath transport bottlenecks. Deployed across Taobao, Eleme, AliCloud, and Dingtalk, serving hundreds of millions of users.

GSO-Simulcast: Led the design and development of a global stream orchestrator deployed at full scale in Dingtalk video conferencing, improving the overall user satisfaction score by 7%.

Standardization: Co-author of the IETF Multipath QUIC draft, adopted as a working group draft at IETF 112 and progressing toward RFC publication.

MIT Media Lab Cambridge, MA, Sept. 2016–Aug. 2018
Postdoctoral Associate

My research focused on Internet-of-Things (IoT)-based human-machine interfaces. I used tools from computer networks, signal processing, machine learning, and hardware design to uncover, analyze, and engineer the networks of natural and man-made signals that fill our world in order to enhance our abilities in communication, sensing, and actuation.

EDUCATION

Ph.D. in Electrical and Computer Engineering
Cornell University (2010–2016).
GPA: 4.21/4.3
B.S. in Electrical Engineering
University of Science and Technology of China (USTC) (2006–2010).
GPA: 4.00/4.3

SELECTED HONORS AND AWARDS

Uber Re-imagine Award finalist, 2024
Keynote speaker at APNET'22 and APNET'21
Keynote speaker at MobiHoc'21
MobiCom'21 Best Paper Runner-Up
DAMO Award Individual finalist (the highest honor in DAMO Academy; Top 0.4%)
Top-10 Breakthrough News story by RFID World, 2020
Covered by IEEE Signal Processing Magazine Special Report, 2018
ACM SIGCOMM research paper highlights, 2018
China Young Scientist Computing Association Rising Star, 2018
Qualcomm QualStar Award, 2015
International Microwave Symposium Best Student Paper Award, 2015
Irwin and Joan Jacobs Scholar, 2011
National Scholarship (Top 2%), Ministry of Education, China, 2009

IETF DRAFTS

Y. Liu, Y. Ma, Q. D. Coninck, O. Bonaventure, C. Huitema, M. Kuhlewind, *Multipath Extension for QUIC*, IETF WG Draft, 2023. **Working group draft in the IETF QUIC WG.**
Z. Zheng, Y. Ma, Y. Liu, M. Kuhlewind, *QUIC-enabled Service Differentiation for Traffic Engineering*, IETF Draft, 2023.
F. Yang, Y. Liu, Y. Ma, *A Configurable Retransmission Extension for HTTP/3 Datagrams*, IETF Draft, 2023.
Y. Ma, Y. Liu, C. Huitema, X. Yu, *An Advanced Scheduling Option for Multipath QUIC*, IETF Draft, 2023.

- Y. Ni, Z. Zheng, X. Lin, F. Gao, X. Zeng, Y. Liu, G. Yang, Y. Su, D. Cai, H. Liu, C. Xu, E. Zhai, Y. Ma *corresponding author*, *CellFusion: Multipath Vehicle-to-Cloud Video Streaming with Network Coding in the Wild*, ACM SIGCOMM 2023. *Cellfusion is a new AliCloud product designed for self-driving customers.*
- B. Wu, K. Qian, B. Li, Y. Ma, Q. Zhang, Z. Jiang, J. Zhao, E. Zhai, D. Cai, X. Jin, *XRON: A Hybrid Elastic Cloud Overlay Network for Video Conferencing at Planetary Scale*, ACM SIGCOMM 2023.
- X. Lin, Y. Ma *corresponding author*, J. Zhang, Y. Cui, J. Li, S. Bai, Z. Zhang, D. Cai, H. Liu, M. Zhang, *GSO-Simulcast: Global Stream Orchestration in Simulcast Video Conferencing*, ACM SIGCOMM 2022. *Deployed in Dingtalk, serving over 500 million users worldwide.*
- Z. Zheng, Y. Ma *corresponding author*, Y. Liu, F. Yang, Z. Li, Y. Zhang, J. Zhang, W. Shi, W. Chen, D. Li, Q. An, H. Hong, H. Liu, and M. Zhang, *XLINK: QoE-driven Multi-path QUIC Transport in Large-scale Video Services*, ACM SIGCOMM 2021. *Deployed in Taobao and successfully spun off into a cloud product.*
- X. Fan, L. Shangguan, S. Rupavatharam, Y. Zhang, J. Xiong, Y. Ma, and R. Howard, *HeadFi: Bringing Intelligence to All Headphones*, ACM MobiCom 2021. *Best paper runner-up.*
- R. Zhao, P. Wang, Y. Ma *corresponding author*, P. Zhang, H. Liu, X. Lin, X. Zhang, C. Xu, and M. Zhang, *NFC+: Breaking NFC Networking Limits through Resonance Engineering*, ACM SIGCOMM, 2020. *Selected as one of RFID World's Top-10 Breakthrough News stories.*
- X. Fan, L. Shangguan, R. Howard, Y. Zhang, Y. Peng, J. Xiong, Y. Ma, X.-Y. Li, *Towards Flexible Wireless Charging for Medical Implants Using Distributed Antenna System*, ACM MobiCom, 2020.
- Z. Luo, Q. Zhang, Y. Ma, M. Singh, and F. Adib, *3D backscatter localization for fine-grained robotics*, USENIX NSDI, 2019. *Featured on MIT News, RFID Journal, Engadget, ZDNet, and other media outlets*
- U. Ha, Y. Ma, Z. Zhong, T. Hsu, and F. Adib, *Learning food quality and safety from wireless stickers*, ACM HotNets, 2018. *Story covered by CBS This Morning*
- Y. Ma, Z. Luo, C. Steiger, G. Traverso and F. Adib, *Enabling deep-tissue networking for miniature medical devices*, ACM SIGCOMM, 2018. (acceptance rate: $40/222 = 18\%$) *Featured on Technology Review, Boston Herald, Engadget, and other media outlets*
- Y. Ma, N. Selby, and F. Adib, *Minding the billions: ultra-wideband localization in deployed RFID tags*, ACM MobiCom, 2017. (acceptance rate: $35/186 = 19\%$)
- Y. Ma, N. Selby, and F. Adib, *Drone relays for battery-free networks*, ACM SIGCOMM, 2017. (acceptance rate: $36/250 = 14\%$) *Featured as a Spotlight on the MIT homepage, IEEE Spectrum, The Verge, Sina, Sohu, and other media outlets*
- Y. Ma, N. Selby, M. Singh, and F. Adib, *Demo: fine-grained RFID localization via ultra-wideband emulation*, ACM SIGCOMM, 2017.
- Y. Ma, X. Hui, and E. C. Kan, *3D real-time indoor localization via broadband nonlinear backscatter in passive devices with centimeter precision*, ACM MobiCom, 2016. (acceptance rate: $32/226 = 14\%$)

MEDIA COVERAGE	CBS This Morning, The Verge, IEEE Spectrum, MIT News (front page), International Business Times, Boston Herald, Technology Review, Digital Trends, Geek, Inverse, Engadget, UPI, RFID Journal, Cornell News, ScienceDaily, Sina (Chinese), Sohu (Chinese), IT Home (Chinese), The Drive, Material Handling and Logistics, RF Global Net, Computer Welt (Austria), Electronics360, Quartz, New Atlas, Design Products and Applications.
INVITED REVIEWER FOR	IEEE Transactions on Mobile Computing IEEE/ACM Transactions on Networking IEEE Consumer Electronics Magazine
OTHER SERVICE	USENIX NSDI Technical Program Committee 2027 ACM MobiCom Technical Program Committee 2025 ACM SIGCOMM Technical Program Committee 2024 ACM HotMobile Technical Program Committee 2024 IEEE GlobeCom Technical Program Committee 2021 IEEE INFOCOM Technical Program Committee 2020 IEEE INFOCOM Technical Program Committee 2019 ACM CoNEXT Technical Program Committee 2018 ACM MobiCom Workshop Technical Program Committee 2018 ACM Mobisys Workshop Technical Program Committee 2018